

Laboratory testing of Andes virus (*Orthohantavirus andesense*) infection

Interim guidance

15 May 2026



Key points

- Confirmation of Andes virus (ANDV) infection is based on RT-PCR (reverse transcription polymerase chain reaction) performed on whole blood to detect viral RNA.
- Hantavirus-wide RT-PCR followed by sequencing can be used to confirm viral species when ANDV-specific assays are unavailable.
- Immunoglobulin M (IgM) enzyme-linked immunosorbent assays (ELISA) or Indirect immunofluorescence assay (IFA) can also confirm diagnosis in acute disease, provided that potential cross-reactivity with co-circulating hantaviruses is considered depending on local context.
- In light of the high mortality of the disease, repeat testing should be considered 24-48 hours later if initial testing is negative but clinical suspicion remains high.
- When handling biological specimens, appropriate safety measures should be used based on the local risk assessment.

Purpose and target audience

This document provides interim guidance for laboratories, clinicians, health workers, public health officials and other stakeholders responding to outbreaks of Andes virus in non-endemic areas (i.e. areas where ANDV is not enzootic in rodent reservoir) and involved in the diagnosis and care of patients with suspected or probable Andes virus, from specimen collection through to diagnostic analysis and reporting.

Introduction

Hantaviruses are a diverse group of RNA viruses with a wide geographical distribution that can be transmitted to humans via contact with rodent excreta. They are classified within the class *Bunyaviricetes*, order *Elliovirales*, family *Hantaviridae*, and genus *Orthohantavirus* (1). They are enveloped, negative-strand RNA viruses whose genome is composed of three segments: the small (S) segment encoding the nucleocapsid (N) protein, the medium (M) encoding the envelope glycoproteins (Gc & Gn) and the large (L) segment encoding the RNA-dependent RNA polymerase (1,2). Hantavirus infectious particles are pleomorphic, surrounded by a lipid envelope, and measure typically about 80-160 nanometers in diameter. They present tetrameric Gn/Gc protein spikes on their surface that mediate viral attachment and entry into host cells. (3)

Orthohantaviruses persist in rodents and can cause two distinct clinical syndromes in humans. The haemorrhagic fever with renal syndrome (HFRS), first associated with hantavirus in 1976, is caused by Old World hantaviruses

such as Hantaan virus, Seoul virus, Dobrava-Belgrade virus and Puumala virus, and predominantly occurs in Europe and Asia (4,5). In contrast, Hantavirus pulmonary syndrome (HPS), sometimes referred to as hantavirus cardiopulmonary syndrome, was first described in 1993 and is caused by New World hantaviruses, such as Sin Nombre virus and Andes virus (ANDV) and is primarily reported in the Americas (6–8). There are more than 60 orthohantaviruses described of which around 25 are recognized as zoonoses (9).

The identified natural reservoir for ANDV is the long-tailed pygmy rice rat, *Oligoryzomys longicaudatus*, although other sigmodontine rodents may occasionally harbour the virus (10). These rodents maintain persistent asymptomatic infection and shed virus in urine, feces and saliva, which contaminates the surrounding environment (11). The reservoir species is primarily associated with rural and peri-domestic habitats in southern Chile and Argentina, particularly forested areas, shrublands, agricultural zones, woodpiles, barns, sheds, and poorly ventilated cabins where rodents nest or forage (12). Human risk is therefore strongly linked to behaviours that increase exposure to contaminated dust or rodent excreta, including cleaning unused buildings, farming, camping, handling firewood, entering enclosed rural structures, and occupational or recreational activities in endemic wilderness areas. Rarely, transmission can occur via rodent bites.

ANDV is associated with high case fatality rates, frequently reported to be up to 50% (13). These estimates remain strongly context-dependent and may vary according to outbreak setting, surveillance intensity, early diagnosis and clinical management. According to recent national surveillance reports, Argentina reported 86 confirmed HPS cases in 2025, including 28 deaths, while Chile reported 35 confirmed cases, including seven deaths. These relatively limited case numbers largely reflect the fact that spillover transmission from infected natural rodent reservoirs to humans remains uncommon (14).

However, unlike other hantaviruses, ANDV is sometimes transmitted from person to person, usually in close proximity settings (15). The first documented evidence of human-to-human transmission of ANDV emerged during the 1996 outbreak, providing epidemiological indications, based on incubation period estimations, compatible with three to four sequential transmission events and supported by genomic data (15). A study in Argentina reported several clusters between 2001 and 2014, all with evidence of human-to-human transmission (11). In Chile, a study described an outbreak in 2011 involving five cases including the index case, two household contacts and two healthcare workers (13). Another outbreak in 2018 in Chubut province, Argentina has caused 34 confirmed infections and 11 deaths after a supposedly single introduction of ANDV from a rodent reservoir into the human population, followed by human-to-human transmission mainly driven by three symptomatic persons who attended crowded social events, highlighting the potential for superspreading events to occur (17). In addition, mother-to-child transmission has been documented (18). Overall, these studies included household exposure, intimate contact, caregiving without suitable personal protective equipment, and prolonged exposure in poorly ventilated or crowded settings. However, it is important to stress that there is currently no evidence for efficient community transmission of ANDV like that observed with highly transmissible respiratory viruses.

The incubation period of hantavirus infection in humans varies by clinical syndrome and causative species. For Andes virus, the incubation period from exposure to symptom onset ranges from 1 to 6 weeks, with a median of 18 days (16). HPS caused by ANDV progresses through four stages (17–20):

- a 2–8-day prodromic phase marked by nonspecific symptoms such as fever, myalgia, headache and gastrointestinal complaints during which patients often seek care and diagnosis can be confirmed during this stage in a proportion of patients; IgM are typically positive at symptom onset or shortly thereafter and Immunoglobulin G (IgG) follow around 3–7 days after symptom onset
- a cardiopulmonary stage, where a dry cough can rapidly progress to severe respiratory distress and shock, with fatality in up to 50% of cases;
- a diuretic stage entered by survivors, during which lung and cardiac function normalize and most patients are discharged within 1–2 weeks without sequelae;

- a convalescent stage, where fatigue and malaise may persist for weeks or months.

Common clinical features of HPS caused by ANDV include high fever, cough, shortness of breath, bilateral pneumonia, headache, abdominal and back pain, along with laboratory findings such as leukocytosis, thrombocytopenia, elevated creatinine, proteinuria and haematuria; however, early symptoms are nonspecific and overlap with other febrile illnesses, making laboratory confirmation essential.

Differential diagnoses, especially during the prodromal phase, should consider the epidemiological profile of the region and can include, among others:

- leptospirosis,
- COVID-19 and viral pneumonia
- atypical pneumonia,
- sepsis with acute respiratory distress syndrome,
- endocarditis with pulmonary oedema,
- influenza,
- pneumonic plague,
- dengue.
- yellow fever
- arenavirus infections

Because ANDV differs from other hantaviruses in its ability to spread between humans, this guidance focuses on testing recommendations for HPS caused by ANDV. It summarizes key considerations for laboratory and diagnostic testing of suspected and probable cases, including appropriate methods and their use in both endemic and non-endemic settings. WHO will continue to review new evidence and update this guidance as needed.

Goal and objectives of testing

Laboratory testing of suspected or probable Andes virus cases should be performed with a goal of informing clinical management and stopping human-to-human transmission via rapid identification, case management, and containment of outbreaks through contact tracing and targeted public health interventions.

Test results may be used to achieve one or more of the following objectives, depending on the local epidemiological context:

- to confirm or exclude a diagnosis of ANDV infection.
- to guide the appropriate clinical management and infection control measures and support contact tracing.
- to monitor the geographic spread and transmission patterns as part of outbreak control and surveillance.
- to identify zoonotic reservoirs of the virus (in combination with epidemiological investigations).
- to track mutations in the virus that could impact transmissibility, severity of the disease, or performance of diagnostic tests.

Indications for Andes virus testing and differential diagnosis

Any individual, meeting the WHO operational outbreak case definition for a suspected or probable case of Andes virus, should be tested. Table 1 reports case definitions from the WHO contact management guidance (21).

Table 1: WHO operational outbreak case definitions of a suspected, probable and confirmed ANDV case

WHO operational outbreak case definitions of a suspected, probable and confirmed ANDV case for suspected human-to-human transmission	
Suspected case	Anyone who shared or visited a conveyance, where there has been a confirmed or probable ANDV case OR had contact with a probable or confirmed ANDV case AND with any acute (or history of) symptoms compatible with ANDV infection, including fever (38°C or above), myalgia, chills, acute gastrointestinal (e.g. nausea, vomiting, diarrhoea, abdominal pain) or acute respiratory (e.g. cough, shortness of breath, chest pain, difficulty breathing) symptoms
Probable case	A person with signs and symptoms of a suspected case that has been evaluated by a health professional AND has a known epidemiological link with a confirmed or probable ANDV case AND for whom laboratory testing was not possible
Confirmed case	A person with laboratory confirmation of ANDV through RT-PCR and/or by serology ¹
Non-case	A suspected or probable case who tests negative for ANDV by RT-PCR and/or serology ²
Contact	A person who was exposed to a confirmed or probable case of ANDV while the case was infectious, through interactions consistent with exposure to respiratory secretions, saliva, blood or other bodily fluids; including • <u>Direct physical contact</u>, including exposure to saliva or other bodily fluids (e.g. care giving, intimate contact, sharing a bed, etc) • <u>Close proximity exposure</u>, defined as being within 2 meters for a cumulative period of more than 15 minutes (e.g. face to face interactions, shared meals or other social gatherings) • <u>Exposure in shared indoor spaces</u> (e.g. shared households, workplaces, healthcare facilities, social gatherings, or travel/conveyance exposure, etc.) • <u>Unprotected exposure in healthcare settings</u>, particularly during patient care, including during aerosol-generating medical procedures, as well as laboratory exposure

Testing of asymptomatic high-risk contacts

Testing blood specimens of pre-symptomatic contacts for ANDV may, in some cases, detect infection before symptom onset; however, results do not predict infectiousness (22). Argentinean and Chilean experts reported in WHO expert consultations that all patients who tested positive prior to symptom onset later developed symptoms.

Asymptomatic testing of high-risk contacts may therefore be considered if resources allow, to 1. ensure cases are recognized and provided supportive treatment as early as possible; 2. to reduce the likelihood of transmission, if any, provided results are interpreted cautiously and communicated clearly. However, testing should NOT be used to determine the end of the follow-up period, which remains fixed at 42 days (6 weeks) after last exposure, regardless of test results. There is limited evidence to support a specific testing interval (i.e. weekly, twice a week,

¹ In settings where new world hantaviruses co-circulated, potential cross-reactivity should be considered when interpreting the results.

² Non-cases who develop symptoms compatible with the suspected case definition after a negative test and within the maximum incubation period after last exposure to a probable or confirmed case should be retested and reclassified as appropriate.

or more). Considerations on testing frequency must balance acceptability of repeated testing among contacts, operational usefulness, and efficient use of available resources.

Testing of asymptomatic contacts could provide data on the incubation period of the disease, and the timing of test positivity, virus shedding, and progression to symptom development. These data would support the risk assessment in subsequent outbreaks, and the ANDV natural history study protocol developed by the WHO Collaborative Open Research Consortium, which is being used as a multi-country effort to gather such data.

Table 2: Considerations for testing asymptomatic contacts

	Justification and evidence	Value
Advantages		
Early identification of infection before symptoms	Scientific studies report detection of ANDV RNA during the pre-symptomatic phase (i.e. before symptoms onset) (22).	Early viral detection in asymptomatic individuals could allow: 1. Anticipatory and early clinical care; and 2. Rapid isolation of the patient to limit further potential transmission
Opportunity to generate scientific knowledge	There is limited data on ANDV pre-symptomatic viraemia, transmission dynamics and the natural history of the disease	Additional data would support evidence-based assessment and guidance development for ANDV infection management and allow for a better understanding of ANDV dynamics to guide management and testing during future ANDV outbreaks. WHO Collaborative Open Research Consortium is coordinating a multi country effort to standardize natural history protocols
Disadvantages		
A negative RT-PCR test result does NOT exclude infection to develop	RT-PCR depends strongly on timing, and viral load may be below the limit of detection during pre-symptomatic disease.	There is a risk of false reassurance when an individual receives a negative RT-PCR result and assumes they are not infected. However, a negative result may simply indicate that the test was performed before the viral load reached the detection threshold, rather than confirming absence of infection. This may reduce adherence to symptom monitoring or delay care-seeking, thereby increasing the risk of onward transmission to contacts.
Unclear link between RT-PCR positivity and the ability to transmit the virus	Detection of ANDV RNA in a clinical sample does not necessarily indicate that the virus can be transmitted to another	Regarding human-to-human transmission, evidence suggests transmission mainly occurs from close and prolonged contact, with

	person, because: 1. RT-PCR detects viral RNA but does not confirm or refute the presence of infectious virus particles, 2. the sample matrix (e.g. blood vs. respiratory discharge) influences the likelihood of transmission.	virus mainly associated with symptomatic (acute) disease (14). RT-PCR positivity in the context of recovery from infection may precipitate prolonged isolation in an individual who does not necessarily pose a risk of onward infection.
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Specimen collection, shipment and storage

Preparation of materials. All required materials should be assembled and ready for use prior to specimen collection. Required materials and equipment may include transport containers, specimen collection bags, triple packaging, coolers and cold packs or dry ice, sterile blood-drawing equipment (e.g. needles, syringes and tubes), labels and permanent markers, personal protective equipment (PPE), and materials for surface decontamination.

Safety procedures. Use of adequate standard operating procedures (SOPs) and rigorous adherence to infection prevention and control guidelines must be ensured during specimen collection and handling (23). Laboratory personnel must be trained for appropriate donning and doffing of PPE, and specimen collection, storage, packaging and transport. Competency assessments should be performed following trainings, and only competent personnel should be authorized to work with the specimens.

Specimen to be collected. The recommended specimen type for laboratory confirmation of ANDV using nucleic acid amplification techniques (RT-PCR) is whole blood. While whole blood is preferred, serum, or plasma may also be used. Buffy coats can also be used, as ANDV demonstrates a strong tropism for mononuclear cells. Serum and plasma are suitable for serological analysis, including ELISA, IFA, or other validated immunoassays (Table 3). Additional specimen types such as urine, saliva/gingival swabs in viral transport media (VTM), semen, cerebrospinal fluid (CSF), lung, spleen, liver biopsies can be considered (Table 3) (24).

Packaging and shipment of clinical specimens. Correct handling and storage of specimens during transportation is essential for accurate diagnostic testing. Transport of specimens should comply with any applicable national and/or international regulations, including the UN Model Regulations and any additional requirements specific to the mode of transport. For international transport, specimens from suspected, probable, or confirmed Andes virus cases should be transported as Category A, UN2814 “infectious substance, affecting humans” (25). ANDV isolates and cultures should also be transported as Category A, UN2814, and shipping requires a dangerous goods-certified shipper (25). Clinical diagnostic specimens may be classified as Category B with the optional use of Category A packaging instruction P620 as an added precaution. All specimens being transported should be packaged, labeled, and accompanied by appropriate documentation in accordance with applicable national and international regulations.

Specimen storage. Whole blood, clotted blood, serum, or plasma can be stored at 2 to 8°C if shipment to the testing laboratory can be completed within 24 hours. Ice packs should be used, if necessary, to maintain these temperatures. For CSF and serum, if a delay in transport to the testing laboratory is anticipated for more than 72 hours, freeze specimens at -80°C when possible (or at equivalent ultra-low temperature) and ship on dry ice. Do not ship clotted blood or EDTA/ heparin/citratd whole blood on dry ice. Repeated freeze-thaw cycles should be avoided as these can degrade specimen quality.

Biological risk management

It is recommended that all manipulation of specimens originating from suspected, probable or confirmed cases of ANDV in the laboratory are conducted according to a risk-based approach. Each laboratory should conduct a local risk assessment for all laboratory activities with consideration for national requirements.

Sample inactivation prior to testing using an inactivation protocol that does not negatively affect diagnostic test performance should be considered to reduce biological risk and biosafety requirements for sample testing.

As ANDV is suspected to be transmitted via aerosols, personnel should avoid procedures that may generate aerosols. When laboratory activities likely to generate aerosol are unavoidable, appropriate controls must be put in place, including the use of appropriate PPEs to protect against exposure.

Laboratory personnel may be exposed to ANDV during specimen processing as a result of contaminated materials or laboratory incidents. When manipulating biological specimens, effective risk-control measures must be applied - indicatively equivalent to biosafety level 2 or 3) as determined by the local risk assessment (29). The following measures should be applied for the purpose of clinical testing without virus propagation:

- Laboratory personnel must be trained appropriately for the safe handling of viruses requiring heightened biosafety measures such as ANDV.
- Specimens from persons who may have ANDV must be handled in a functioning and certified biosafety cabinet (Class I, II or III) prior to specimen inactivation. Properly inactivated specimens do not require a biosafety cabinet.
- Each laboratory should ensure that local inactivation protocols have been validated. Chemical inactivation with methanol, paraformaldehyde, acetone/methanol and detergent-containing lysis buffer as well as physical forces such as ultraviolet irradiation and filtration have been shown to effectively reduce viral infectivity in infected cells and supernatants below detectable levels (26).
- Laboratory personnel should wear appropriate personal protective equipment, particularly when handling specimens prior to inactivation, as established protocols for handling suspected, probable, confirmed ANDV suggest (14).
- All waste that may contain Andes virus should be disinfected before disposal using validated methods, such as autoclaving or chemical disinfection.
- Effective disinfectants include detergent-containing lysis buffers (27).
- Additional control measures should be considered for specific procedures, including aerosol-generating procedures, based on the local risk assessment.
- More information on heightened control measures can be found in the fourth edition of the WHO Biosafety Manual (28).

Laboratory testing methods

ANDV infections have complex infection dynamics. Results must be interpreted alongside clinical presentation and epidemiologic exposure. The incubation period ranges from 1–6 weeks (median ~18 days). By symptoms onset, viremia is usually high, and IgM antibodies are often already detectable, while IgG appear shortly thereafter. IgM levels typically wane within ~1-3 months, while IgG antibodies persist for years. A laboratory testing algorithm for clinical diagnosis and surveillance of ANDV using RT-PCR and serological assays is described in Figure 1. It is important to note that in non-endemic areas, ANDV-specific tests are not regulatory approved, nor have they been validated. In such emergency response situations, laboratories can rely on their own quality systems for quality purposes (e.g. ISO15189).

Nucleic acid amplification testing (RT-PCR)

WHO recommends that confirmation of ANDV infection is achieved using a nucleic acid amplification test (NAAT), specifically laboratory-based reverse transcriptase polymerase chain reaction (RT-PCR), using real-time or conventional PCR on whole blood for the detection of unique sequences of ANDV RNA. Serum, plasma or buffy coats can also be used. When ANDV-specific RT-PCR assays are not available (30), an hantavirus-generic RT-PCR with primers from conserved genomic regions (e.g. S and/or L segment) may be used, followed by sequencing to confirm ANDV (31). The M segment is less frequently targeted as it is less conserved than the S and L segments. When specimens are collected from suspected patients at the appropriate time point at the onset of viremia, this method can yield accurate results (31).

However, one (or more) negative results should not rule out hantavirus infection in a symptomatic patient. Given the lethality of HPS, a negative RT-PCR test may be followed by repeat testing 24-48 hours after initial test when the index of suspicion is high. Several factors could cause a negative result in an infected individual, including:

- Poor quality of the specimen containing little patient material (as a control, consider determining whether there is adequate human DNA in the sample by including a human target in the PCR testing).
- The specimen was not handled and shipped appropriately.
- The specimen was collected late or very early in the infection.
- Technical reasons inherent in the test, e.g., virus mutation or PCR inhibition.

Each RT-PCR run should include appropriate external and internal controls, and laboratories are encouraged to implement a robust laboratory quality management system and to participate in external quality assessment (EQA) schemes when available. It is also recommended that laboratories order their own primers and probes to perform entry testing/validation to assess assay performance and detect potential contamination.

In-house protocols that have undergone clinical and analytical evaluation are available and the corresponding primer and probe sequences can be found in published articles(31,32). However, commercially available research use only kits are also available for purchase. When selecting commercial kits, ensure they can detect the circulating virus of interest, as most available kits target old-world hantavirus strains. Such tests may be accredited but are most likely not fully validated, and the commercial test kits used may not be regulatory approved. In such emergency response situations, laboratories can rely on their own quality systems for quality purposes (e.g. ISO15189). Although not specific to zoonotic viruses such as hantavirus, WHO has published a template protocol titled “First few X cases and contacts diagnostic test evaluation for respiratory pathogens with pandemic potential” that countries can adapt to validate their assays (33).

Serological testing

Typically, most cases of acute HPS elicit the production of IgM antibodies on fever onset, and IgG shortly thereafter, making serological tests detecting antibodies to viral antigens a standard and reliable approach for laboratory diagnosis of recent or past infections (5). One major setback of serological tests is cross-reactivity with other new world hantavirus species (e.g. Sin Nombre virus), and sometimes, serology in general can be subject to cross-reactivity with other common viruses (e.g. Epstein-Barr virus) or interference material (eg. Rheumatoid factors). As sero-prevalence studies have shown, exposure to hantavirus has been detected in many regions around the globe, serological results must be interpreted with caution, especially IgG results in areas without prior reported cases or exposure or areas endemic for cross-reacting orthohantaviruses (34,35). Of note, several orthohantavirus strains are yet to be taxonomically assigned, complicating the interpretation of serological results in such contexts.

Laboratories can leverage the technical capacities and resources in serology, such as equipment established for diseases like measles and rubella, to establish testing capacity for hantavirus. Moreover, several commercial kits for both old-world and new-world hantaviruses are available for purchase.

- IgM ELISA/IFA -IgM antibodies against viral antigen epitopes usually develop at symptoms onset, rise (within a week), and later clear after 1-3 months. In the context of the contact tracing efforts for the 2026 cruise-ship outbreak, it is unlikely that individuals have been exposed to other new world hantaviruses, but cross-reactivity with other commonly cross-reacting viruses (e.g. EBV) or interference material (e.g. rheumatoid factors) should also be considered if the test used was not fully validated.
- IgG ELISA/IFA - ELISA for IgG antibodies against hantavirus antigen is also a common option for diagnosing hantavirus infection. IgG antibodies usually develop shortly after IgM, and remain for longer periods (years). While this method can yield high-quality results, it is also not available in the early stages of the disease, when critical management decisions are required. Furthermore, cross-reactivity with other hantavirus strains can result in false positivity in areas where hantaviruses are endemic or co-circulate.

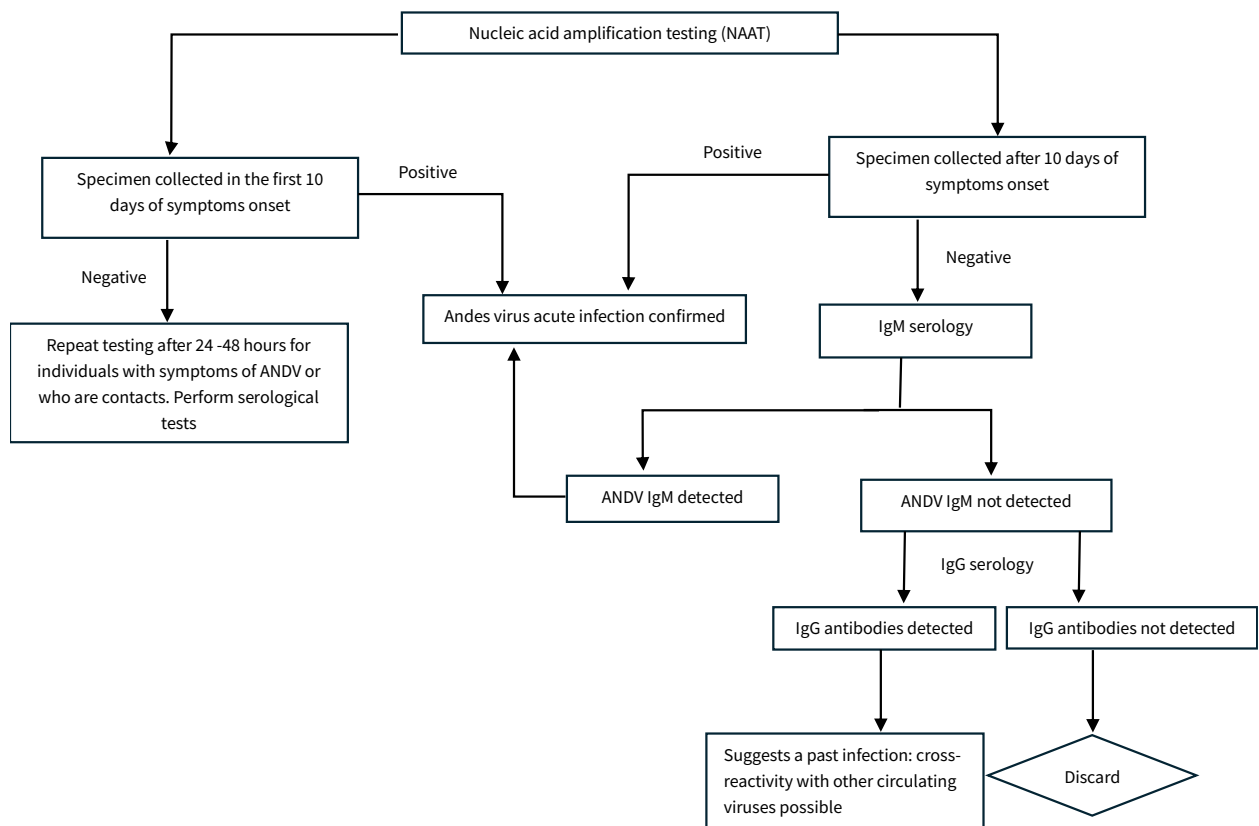


Figure 1: Laboratory testing algorithm for clinical diagnosis and surveillance of Andes Virus using RT-PCR and serological assays. Note that, for diagnostic purposes, IgM-ELISAs or IFAs are used to confirm serology. If deemed appropriate, RT-PCR and serology can also be performed in parallel.

Antigen-based rapid detection tests (RDTs).

Currently, there are no antigen-based RDTs for hantavirus infection that have been developed, evaluated, validated, and approved.

Viral culture.

Virus isolation is not recommended as a routine diagnostic procedure. Isolation of hantaviruses is technically demanding, requires specialized expertise and Biosafety level 3 laboratory facilities. Because of these complexities and as this method is not recommended for routine diagnostic purposes, detailed protocols are not included in this guidance.

Genomic sequencing

Sequencing methods vary widely and can range from Sanger sequencing of a partial genome fragment to whole genome sequencing using Illumina or Oxford Nanopore-based methods. Limited anecdotal data suggests that sequencing directly from human whole blood or sera is more challenging than sequencing total RNA extracted from lung tissue (fresh or formalin-fixed paraffin-embedded) or kidney tissue. Selection of the sequencing method depends on the expected strain diversity for a particular region. Unbiased next generation sequencing can detect unknown viral strains (or hantavirus-like organisms), but at the cost of reduced sensitivity compared to targeted methods. In contrast, targeted sequencing methods, such as viral probe-based enrichment or amplicon-based methods, are more appropriate for specimens with low viral loads. Target-based methods exhibit increased viral sensitivity at the cost of reduced specificity. Enrichment-based methods using a pan-viral probe set appear to balance these trade-offs, enabling enhanced sensitivity while detecting a wide range of hantaviral diversity. Genomic sequencing is also valuable for assessing whether diagnostic primer and probe sets detect the full diversity of endemic hantaviruses.

WHO strongly encourages countries and laboratories to share genomic sequence data in real-time through the available open access databases.

Table 3: Overview of testing methods and use cases

Testing method	Specimen type	Use cases
RT-PCR	Whole blood (preferred); serum, plasma, buffy coats can also be used Other specimens: urine, saliva/gingival swab in VTM, semen, CSF, lung, spleen, liver biopsies	- Confirm acute infection when viremia is present (typically at or near symptom onset). - Differentiate Andes virus from other hantaviruses using sequencing when only generic RT-PCR is available. - Guide early clinical management when rapid, definitive diagnosis is needed before serology becomes fully interpretable. - Negative results do not exclude the diagnosis. Retesting may be required on a case-by-case basis
Serology	Serum (preferred), plasma	- Identify recent infection through IgM and IgG detection - Support diagnosis when RT-PCR is negative later in illness.

		- Serological testing at the beginning and end of the follow up period may also be considered to ascertain serological status of contacts.
Genomic sequencing	Any sample type positive by RT-PCR with sufficient viral load	- Track viral evolution by identifying mutations, lineage changes, and emergence of novel variants. - Support epidemiological investigations through phylogenetic analysis that links cases, clusters, and transmission chains. - Characterize viral properties such as virulence, host adaptation, and potential impacts on diagnostics or therapeutics. - Confirm etiology when pan-hantavirus RT-PCR approaches are employed.
Other (e.g. Immunohistochemistry and histopathological analysis in deceased persons)	Preferred: Blood by cardiac puncture and/or lung tissue puncture Lung tissue or respiratory tract specimen (fresh or preserved in formaldehyde)	- Investigate deceased individuals meeting the suspected or probable case definition via laboratory analyses, following applicable biosafety measures

Interpretation and reporting of testing results

A positive result by species-specific RT-PCR confirms ANDV infection. In the absence of ANDV-specific RT-PCR, hantavirus-generic RT-PCR followed by sequencing can be used to confirm ANDV infection.

For serological tests, An IgM positive result is confirmatory of the clinical diagnosis, but cross-reactivity with co-circulating hantaviruses should be considered based on the local context.

All hantavirus test results, positive or negative, should be reported to national authorities immediately in accordance with local requirements. As the confirmed case definition does not require the presence of symptoms, and in the context of screening of asymptomatic contacts, a PCR positive result should be considered confirmatory and therefore reported as above.

Quality assurance: key considerations

Positive control material for RT-PCR assays can be ordered from specialized initiatives such as European Virus Archive (<https://www.european-virus-archive.com/>). As best practice, positive controls should be included at a low concentration which is unequivocally above the limit of detection. Inclusion of quality control materials, where possible, can assist in controlling any assay issues. Controls should provide information about (1) specimen quality, (2) nucleic acid quality, and (3) process quality. Because RT-PCR (particularly if nested protocols are employed) can be extremely sensitive, efforts should be made to avoid contamination, and negative controls should be used on every run to ensure contamination has not occurred. Specimen integrity controls (e.g. RNase

P), and extraction, positive and inhibition controls can help in distinguishing a false negative from a true negative. Controls should be used following laboratory SOPs. If any of the assay controls fail, testing should be repeated. Where possible, participation in external quality assessment for molecular detection of hantaviruses is recommended. Quality assurance should be ensured in accordance with national or international quality systems (e.g. ISO 15189:2022).

Global laboratory networking

Access to timely and accurate laboratory testing of specimens from ANDV suspected cases and contacts under investigation is an essential part of the diagnosis and surveillance of ANDV infections. Countries expecting imported cases or contacts should strengthen diagnostic capacity at all levels of the health system and, at a minimum, should have access to reliable testing at national level or through referral to laboratories in other countries that are willing and able to perform hantavirus diagnostics. Diagnostic Network Optimization is critical for an efficient and coordinated diagnostic response. Countries should design integrated sample transportation systems and use existing laboratory capacity across diseases. WHO, through its Regional Offices, can assist Member States in accessing testing and sequencing through referral.

Process and methods for guidance development

This document was prepared in consultation with subject matter laboratory experts with experience handling and detecting ANDV and other hantaviruses, as well as individuals with expertise in public health virology and the development of diagnostic assays for hantaviruses.

In May 2026, the WHO Secretariat convened an expert panel to review draft recommendations to include in the interim guidance related to laboratory testing of ANDV infection.

Stepwise approach

Step 1: Define key questions

The WHO Secretariat held preparatory conference calls with key expert groups from various countries to identify and list key questions for which there was a need to review the latest evidence related to Andes virus diagnostics, laboratory biosafety, specimen handling, and surveillance strategies. The Secretariat also reviewed and revised existing interim guidance documents for hantavirus outbreak preparedness and response to ensure relevant updates would be covered.

Step 2: Review evidence.

A search using one search string for each question was performed online via PubMed by the WHO Secretariat. This scoping review focused on updating evidence related to the performance of molecular assays, serological methods, and emerging near-patient or point-of-care diagnostic tools for Andes virus and other hantaviruses. Additional research was conducted in other relevant sources, including reports from reference laboratories, regional hantavirus surveillance networks, and manufacturers of diagnostic assays.

The limitations of this review include the absence of a formal GRADE process due to the accelerated timeline, the absence of support from a methodologist, preventing the systematic use of PICO questions, comprehensive systematic reviews, and detailed consideration of values, preferences, costs, resources, and feasibility. Additionally, a systematic assessment of risk of bias was not conducted, which may affect the reliability of the findings.

Step 3: Guidance drafting and review

WHO drafted a guidance that was sent for review by a multidisciplinary panel of virologists, scientists, public health officials, clinicians, and laboratory specialists with experience in the diagnosis and management of hantavirus infections and other emerging zoonotic diseases. On 12 May 2026 a meeting was held to discuss key points of the guidance with the same group of experts. All external contributors completed a Declaration of Interest. No conflict was identified.

Step 4: Review updated version.

WHO updated the interim guidance to incorporate feedback from experts and circulated the updated version to the expert group, responsible officers involved in outbreak preparedness and response activities, and regional laboratory focal points working on hantavirus surveillance and diagnostics.

Step 5: Publish and disseminate.

The final document was submitted and approved for WHO executive clearance prior to publication and dissemination.

Plans for updating

This interim guidance incorporates the latest understanding and characteristics of ANDV and addresses questions and issues received from WHO Country and Regional offices and other channels. WHO continues to monitor the situation closely for any changes that may affect this document. Should any factors change, WHO will issue a further update in six months time.

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Abbreviations

ANDV	Andes virus
CSF	Cerebrospinal Fluids
DNA	Deoxyribonucleic acid
ELISA	Enzyme-linked immunosorbent assay
EQA	External quality assessment
HFRS	Haemorrhagic fever with renal syndrome
HPS	Hantavirus pulmonary syndrome
IFA	Indirect immunofluorescence assay
IgG	Immunoglobulin G
IgM	Immunoglobulin M
ISO	International Organization for Standardization
PPE	Personal protective equipment
RDT	Rapid diagnostic test
RNA	Ribonucleic acid
RT-PCR	Reverse transcription polymerase chain reaction

SOPs Standard operating procedures
UN2814 United Nations classification for infectious substances affecting humans
VTM Viral transport medium
WHO World Health Organization

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